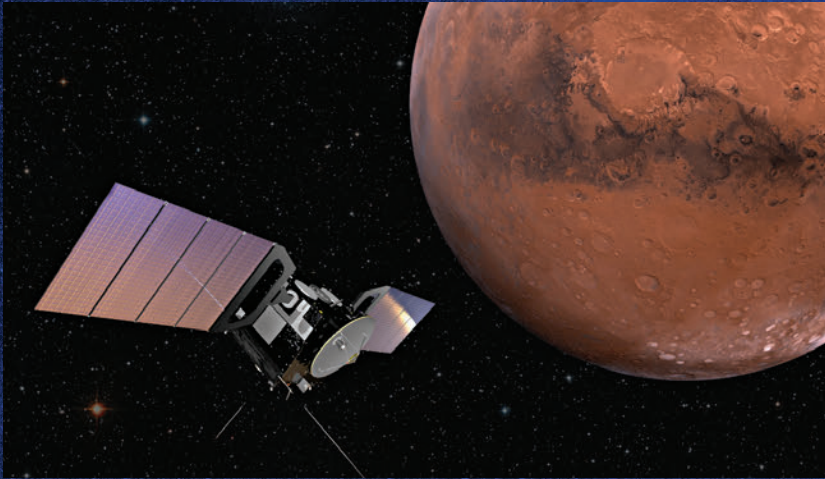




AI maps the weather of Mars

European orbiters catalog 1,000 dust devils

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Mars is often imagined as a quiet, rusty desert, frozen in time. But look closer, and the Red Planet reveals a secret: it's alive with whirling dust devils – miniature tornadoes that dance across its plains, carrying dust, shaping terrain, and hinting at a restless atmosphere. Thanks to two decades of European orbital surveillance and the power of artificial intelligence, scientists have now mapped 1,039 of these elusive Martian twisters, painting a portrait of the planet's hidden winds.

Orbiters and AI

Tracking dust devils from hundreds of kilometers above the surface isn't exactly easy. Enter ESA's Mars Express and the ExoMars Trace Gas Orbiter, two orbiters armed with high-resolution cameras. Over nearly 20 years, these spacecraft

collected thousands of images of the Martian surface. The challenge? Dust devils are fleeting – fast, small, and often invisible to the naked eye.

This is where AI became a cosmic detective. Machine learning algorithms scanned the orbital images, detecting tiny shifts and subtle distortions caused by moving dust col-

umns. These invisible fingerprints revealed the speed, direction, and frequency of Martian dust devils, uncovering winds that sometimes roar at 158 km/h.

Seasonal patterns

The dust devil map revealed more than numbers. Most of the tornadoes appear over dust-laden plains like Amazonis Planitia, and they follow a rhythm: peaking during spring and summer, typically in the late morning and early afternoon.

For robotic explorers, these winds are both a challenge and a blessing. Dust can coat solar panels and slow a rover's progress, but a well-timed Martian dust devil can also act like a natural windshield wiper, cleaning panels and giving machines a boost of energy.

By mapping 1,000 dust devils, scientists now have a global view of Martian atmospheric dynamics, helping predict storm patterns, temperature fluctuations, and the movement of airborne dust.

AI in planetary science

This discovery wouldn't have been possible without AI. Manual analysis of such a massive image archive would have taken decades. Machine learning acted like a supercharged pair of eyes, detecting patterns invisible to humans and accelerating the pace of discovery. **d**

